

Q1. For the relation STUDENT_COURSE(SID, SName, CID, CName, Instructor, Grade), identify all functional dependencies and determine the candidate keys.

STUDENT_COURSE(SID, SName, CID, CName, Instructor, Grade)

Functional Dependencies:

- $SID \rightarrow SName$
- $CID \rightarrow CName$
- $CID \rightarrow Instructor$
- $(SID, CID) \rightarrow Grade$

Candidate Key:

(SID, CID)

Explanation:

SID alone cannot uniquely identify a course because a student can enroll in many courses. CID alone cannot uniquely identify a student because a course can have many students. Therefore, SID and CID together uniquely identify an enrollment.

Q2. Demonstrate the process of converting an unnormalized relation into 1NF with step-by-step justification and example.

Consider the following unnormalized relation:

SID	SName	Courses
101	Arun	DBMS, OS
102	Ravi	CN, DBMS

The Courses attribute contains multiple values in a single cell. This violates the atomicity requirement of 1NF.

Step 1: Identify repeating groups

For example, student 101 has two courses: DBMS and OS.

Step 2: Remove repeating groups

Create a separate row for each course.

SID	SName	Course
101	Arun	DBMS
101	Arun	OS
102	Ravi	CN
102	Ravi	DBMS

Step 3: Justification

Now every attribute contains a single atomic value and there are no repeating groups.

Therefore, the relation is in First Normal Form (1NF).

Q3. Given relation R(A,B,C,D,E) with FDs: $A \rightarrow BC$, $BC \rightarrow D$, $D \rightarrow E$. Analyze and decompose it to 2NF eliminating partial dependencies.

Given:

R(A, B, C, D, E)

Functional Dependencies:

- $A \rightarrow BC$
- $BC \rightarrow D$
- $D \rightarrow E$

Step 1: Find the candidate key

Calculate A^+ :

$$A^+ = \{A\}$$

Using $A \rightarrow BC$:

$$A^+ = \{A, B, C\}$$

Using $BC \rightarrow D$:

$$A^+ = \{A, B, C, D\}$$

Using $D \rightarrow E$:

$$A^+ = \{A, B, C, D, E\}$$

Therefore:

$$A^+ = ABCDE$$

Hence, A is a candidate key.

Step 2: Check partial dependency

A partial dependency occurs when a non-key attribute depends on only part of a composite candidate key.

Here, the candidate key is a single attribute A. Therefore, partial dependency is not possible.

Conclusion:

The relation is already in 2NF.

However, it is not in 3NF because:

$$D \rightarrow E$$

D is not a superkey.

A possible 3NF decomposition is:

- $R_1(A, B, C, D)$
- $R_2(D, E)$

Q4. Apply 3NF decomposition on the relation EMPLOYEE(EmpID, EmpName, DeptID, DeptName, ManagerID) with transitive dependencies.

Relation:

EMPLOYEE(EmpID, EmpName, DeptID, DeptName, ManagerID)

Functional Dependencies:

- $\text{EmpID} \rightarrow \text{EmpName}, \text{DeptID}$
- $\text{DeptID} \rightarrow \text{DeptName}, \text{ManagerID}$

Transitive Dependency

From:

$\text{EmpID} \rightarrow \text{DeptID}$

and

$\text{DeptID} \rightarrow \text{DeptName}, \text{ManagerID}$

we get:

$\text{EmpID} \rightarrow \text{DeptName}, \text{ManagerID}$

This is a transitive dependency.

Decomposition

EMPLOYEE(EmpID, EmpName, DeptID)

Primary Key: EmpID

DEPARTMENT(DeptID, DeptName, ManagerID)

Primary Key: DeptID

The transitive dependency is removed, and the resulting relations are in 3NF.

Q5. Verify whether $R(A,B,C,D)$ with FDs: $AB \rightarrow C$, $C \rightarrow D$, $D \rightarrow A$ is in BCNF. If not, decompose it into BCNF.

Given:

$R(A, B, C, D)$

Functional Dependencies:

- $AB \rightarrow C$
- $C \rightarrow D$
- $D \rightarrow A$

Step 1: Find candidate keys

AB^+ :

$AB \rightarrow C$

$AB^+ = ABC$

$C \rightarrow D$

$AB^+ = ABCD$

Therefore, AB is a candidate key.

BC^+ :

$BC \rightarrow D$

$D \rightarrow A$

Therefore:

$BC^+ = ABCD$

So BC is a candidate key.

BD^+ :

$D \rightarrow A$

$BD^+ = ABD$

$AB \rightarrow C$

$$BD^+ = ABCD$$

So BD is also a candidate key.

Candidate keys are:

AB, BC, BD

Step 2: Check BCNF

For BCNF, the determinant of every non-trivial functional dependency must be a superkey.

FD	Determinant	BCNF
$AB \rightarrow C$	AB is a key	Satisfies
$C \rightarrow D$	C is not a key	Violates
$D \rightarrow A$	D is not a key	Violates

Therefore, R is not in BCNF.

Step 3: Decomposition

Using the violating dependency:

$$C \rightarrow D$$

Decompose into:

- $R_1(C, D)$
- $R_2(A, B, C)$

In R_1 , C is the key.

In R_2 , AB is the key.

Therefore, both relations satisfy BCNF.

Q6. Experimentally verify lossless-join decomposition for a given relation using the tabular (Chase) algorithm.

Consider:

$R(A, B, C)$

Decomposition:

- $R_1(A, B)$
- $R_2(B, C)$

Given FD:

$B \rightarrow C$

Step 1: Initial Tableau

Row	A	B	C
R1	a	b	c ₁
R2	a ₂	b	c

Step 2: Apply $B \rightarrow C$

Both rows have the same value b for B.

Therefore, their C values must become the same.

Row	A	B	C
R1	a	b	c
R2	a ₂	b	c

Step 3: Conclusion

A row contains all the required distinguished symbols.

Therefore, the decomposition is lossless.

Hence:

$$R = R1 \bowtie R2$$

is a lossless decomposition under the given functional dependency.

Q7. Identify the multi-valued dependencies in TEACHER(TID, Subject, Hobby) and decompose it into 4NF.

Relation:

TEACHER(TID, Subject, Hobby)

Suppose a teacher can teach multiple subjects and can have multiple hobbies independently.

The multi-valued dependencies are:

- $TID \twoheadrightarrow Subject$
- $TID \twoheadrightarrow Hobby$

For example:

TID	Subject	Hobby
T1	DBMS	Cricket
T1	DBMS	Music
T1	OS	Cricket
T1	OS	Music

The combinations create unnecessary redundancy.

Since TID is not a superkey of the original relation, the MVD violates 4NF.

Decomposition

TEACHER_SUBJECT(TID, Subject)

TID	Subject
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T1	DBMS
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T1	OS
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TEACHER_HOBBY(TID, Hobby)

	TID	Hobby
T1		Cricket
T1		Music

Thus, the relation is decomposed into 4NF relations.

Q8. Demonstrate how join dependencies lead to redundancy. Decompose a given relation to 5NF (Project-Join Normal Form).

Consider:

R(Supplier, Part, Project)

Suppose suppliers, parts and projects have independent relationships.

Storing all combinations in one relation can create redundancy.

For example:

Supplier	Part	Project
S1	P1	J1
S1	P1	J2
S1	P2	J1
S1	P2	J2

The same supplier-part, supplier-project and part-project information is repeated.

The join dependency can be represented as:

$$R = \pi_{SP}(R) \bowtie \pi_{SJ}(R) \bowtie \pi_{PJ}(R)$$

Decomposition

1. SP(Supplier, Part)
2. SJ(Supplier, Project)
3. PJ(Part, Project)

This decomposition stores the independent relationships separately and eliminates unnecessary redundancy.

Under the stated business rules, the decomposition achieves 5NF (Project-Join Normal Form).

Q9. Given a poorly designed database schema for a college, identify all anomalies (insertion, deletion, update) and normalize it to 3NF.

Consider the poorly designed relation:

COLLEGE(StudentID, StudentName, CourseID, CourseName, InstructorID, InstructorName)

Functional Dependencies:

- StudentID $\rightarrow$ StudentName
- CourseID $\rightarrow$ CourseName, InstructorID
- InstructorID $\rightarrow$ InstructorName
- StudentID, CourseID $\rightarrow$ Enrollment information

Anomalies

1. Insertion Anomaly

A new course cannot be inserted unless at least one student is enrolled in that course.

2. Update Anomaly

If an instructor's name changes, multiple records have to be updated. If one record is missed, inconsistent data occurs.

3. Deletion Anomaly

If the only student enrolled in a course is deleted, information about that course may also be lost.

3NF Decomposition

STUDENT(StudentID, StudentName)

Primary Key: StudentID

COURSE(CourseID, CourseName, InstructorID)

Primary Key: CourseID

INSTRUCTOR(InstructorID, InstructorName)

Primary Key: InstructorID

ENROLLMENT(StudentID, CourseID)

Primary Key: (StudentID, CourseID)

This decomposition reduces redundancy and eliminates the major insertion, deletion and update anomalies.

Q10. Compute the closure of attribute sets for the given FDs and determine all candidate keys for the relation.

Consider:

$R(A, B, C, D, E)$

Functional Dependencies:

- $A \rightarrow B$
- $B \rightarrow C$
- $CD \rightarrow E$
- $E \rightarrow A$

A^+

Start with:

$A^+ = \{A\}$

Using $A \rightarrow B$:

$$A^+ = \{A, B\}$$

Using $B \rightarrow C$:

$$A^+ = \{A, B, C\}$$

$CD \rightarrow E$ cannot be applied because D is not present.

Therefore:

$$A^+ = ABC$$

A is not a candidate key.

$$E^+$$

Start with:

$$E^+ = \{E\}$$

Using $E \rightarrow A$:

$$E^+ = \{E, A\}$$

Using $A \rightarrow B$:

$$E^+ = \{E, A, B\}$$

Using $B \rightarrow C$:

$$E^+ = \{E, A, B, C\}$$

Therefore:

$$E^+ = ABCE$$

E is not a candidate key.

$$CD^+$$

Start with:

$$CD^+ = \{C, D\}$$

Using $CD \rightarrow E$:

$$CD^+ = \{C, D, E\}$$

Using $E \rightarrow A$:

$$CD^+ = \{A, C, D, E\}$$

Using $A \rightarrow B$:

$$CD^+ = \{A, B, C, D, E\}$$

Therefore:

$$CD^+ = ABCDE$$

Hence, CD is a candidate key.

$$BD^+$$

Start with:

$$BD^+ = \{B, D\}$$

Using $B \rightarrow C$:

$$BD^+ = \{B, C, D\}$$

Using $CD \rightarrow E$:

$$BD^+ = \{B, C, D, E\}$$

Using $E \rightarrow A$:

$$BD^+ = \{A, B, C, D, E\}$$

Therefore:

$$BD^+ = ABCDE$$

Hence, BD is a candidate key.

Final Candidate Keys

Candidate Keys = {C,D} and {B,D}

Attribute Set	Closure	Candidate Key
A	ABC	No
E	ABCE	No
CD	ABCDE	Yes
BD	ABCDE	Yes